

1 Preliminary Knowledge

1.1 Quadrature Operator

We define the quadrature operator $\hat{X}_\theta$ as

$$\hat{X}_\theta = \hat{a}e^{-i\theta} + \hat{a}^\dagger e^{i\theta} \quad (1)$$

The conventional field quadratures used in quantum optics are $\hat{X} = \hat{X}_0$ and $\hat{P} = \hat{X}_{\pi/2}$

$$\hat{X} = \hat{a} + \hat{a}^\dagger, \quad \hat{P} = (\hat{a} - \hat{a}^\dagger)/i \quad (2)$$

and satisfy the commutation relation

$$[\hat{X}, \hat{P}] = 2i, \quad (3)$$

building upon the fact that $[\hat{a}, \hat{a}^\dagger] = 1$. These field quadratures define the Wigner function plane.

1.2 Fock States

In the Fock basis representation, Fock (or number) states are denoted as $|n\rangle$, where n is the number of photons (or excitations) in the vacuum $|0\rangle$. They are eigenstates of the number operator

$$\hat{n}|n\rangle = n|n\rangle, \quad (\hat{n} = \hat{a}^\dagger \hat{a}). \quad (4)$$

As eigenstates of the number operator, they have a well-defined number of photons $(\Delta\hat{n})^2 = \langle\hat{n}^2\rangle - \langle\hat{n}\rangle^2 = 0$, but an undefined optical phase ϕ . Their wigner function is a gaussian centered at the origin.

In the position basis representation, a fock state is written as

$$\psi_n(x) = \langle x|n\rangle = \frac{1}{\pi^{1/4}\sqrt{2^n n!}} H_n(x) e^{-x^2/2}, \quad (5)$$

where $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$ is the Hermite polynomial of order n .

1.3 Coherent States

Coherent states are defined as the eigenstates of the annihilation operator

$$\hat{a}|a\rangle = a|a\rangle, \quad a \in \mathbb{C}. \quad (6)$$

The complex amplitude of the state can be written as $\langle\hat{a}\rangle = a = |a|e^{i\phi}$, where ϕ is the optical phase of the electromagnetic field. In the Fock basis, a coherent state is decomposed as

$$|a\rangle = e^{-\frac{|a|^2}{2}} \sum_{n=0}^{\infty} \frac{a^n}{\sqrt{n!}} |n\rangle. \quad (7)$$

The average number of photons in a coherent state is

$$\langle a|\hat{n}|a\rangle = |a|^2. \quad (8)$$

Such state can be generated through displacement of the vacuum

$$|a\rangle = \hat{D}(a)|0\rangle, \quad \hat{D}(a) = \exp[a\hat{a}^\dagger - a^*\hat{a}]. \quad (9)$$

In our convention, the Wigner function of a coherent state is displaced from the origin by an amount $2|a|$ and at an angle ϕ from the x-axis.

1.4 Squeezed States

Squeezed states are generated through squeezing of the vacuum

$$|z\rangle = \hat{S}(z) |0\rangle, \quad \hat{S}(z) = \exp \left[\frac{1}{2} (z \hat{a}^2 - z^* \hat{a}^{\dagger 2}) \right]. \quad (10)$$

The complex number $z = |r|e^{i\theta}$ determines the squeezing parameter r and the squeezed quadrature $\hat{X}_\theta$. Squeezing the quadrature $\hat{X}_\theta$ results in the anti-squeezing of quadrature $\hat{X}_{\theta+\pi/2}$

$$(\Delta \hat{X}_\theta)^2 \propto e^{-2r}, \quad (\Delta \hat{X}_{\theta+\pi/2})^2 \propto e^{2r}. \quad (11)$$

The average photon number of a squeezed state is

$$\langle z | \hat{n} | z \rangle = \sinh^2(r). \quad (12)$$

The Wigner function is elliptical due to the squeezing, and at an angle $\theta/2$ with respect to the x-axis.

1.5 Displaced Squeezed States (DSS)

Such states are generated through squeezing and displacement of the vacuum

$$|a, z\rangle = \hat{D}(a) \hat{S}(z) |0\rangle. \quad (13)$$

Previous points for displaced and squeezed states hold independently. The average number of photons in a DSS is

$$\langle a, z | \hat{n} | a, z \rangle = |a|^2 + \sinh^2(r). \quad (14)$$

The Wigner function is again elliptical, displaced from the origin by an amount $2|a|$, is at an angle ϕ with respect to the x-axis, and rotated by $\theta/2$.

2 Discrimination between pure Quantum States in a communication channel

The minimum theoretical error probability of mis-discriminating between two non-orthogonal pure states $|\psi_{1,2}\rangle$ (with equal prior probability of being selected) is given by the Helstrom bound

$$P^{(\min)}(|\psi_1\rangle, |\psi_2\rangle) = \frac{1}{2} \left(1 - \sqrt{1 - |\langle \psi_1 | \psi_2 \rangle|^2} \right) \quad (15)$$

Experimentally, one cannot reach the absolute minimum probability of making an error when trying to discriminate between two states. In this work, Homodyne detection is considered as the experimental means of a receiver to discriminate between two states, which clearly leads to a higher error probability. The strategy used by the receiver is the following:

$$\begin{cases} x \geq 0 \Rightarrow |0\rangle \equiv |\psi_1\rangle, \\ x < 0 \Rightarrow |1\rangle \equiv |\psi_2\rangle \end{cases} \quad (16)$$

where x is drawn from the Homodyne detection probability distribution

$$x \sim P^{(\text{HD})}(x; |\psi_{1,2}\rangle) = |\langle x | \psi_{1,2} \rangle|^2, \quad (17)$$

with $\hat{X}|x\rangle = x|x\rangle$. The probability of making an error is given by

$$P_{\text{err}} = \frac{1}{2}[p(x \geq 0|x=1) + p(x < 0|x=0)] = p(x \geq 0|x=1); \quad p(x \geq 0|x=1) = p(x < 0|x=0) \quad (18)$$

The conditional probability is given by

$$p(x \geq 0|x=1) = \int_0^\infty P^{(\text{HD})}(x; |\psi_{1,2}\rangle) dx. \quad (19)$$

To approximate this probability, we draw thousands of samples from $P^{(\text{HD})}(x; |\psi_{1,2}\rangle)$, and define the error probability as

$$p(x \geq 0|x=1) = P_{\text{err}} = \frac{\text{\# of samples where } x \geq 0}{\text{\# of all samples}} \quad (20)$$

$$p(x < 0|x=0) = P_{\text{err}} = \frac{\text{\# of samples where } x < 0}{\text{\# of all samples}} \quad (21)$$

The goal of the work is to determine whether the application of certain combinations of quantum resources on the states (e.g. displacement, squeezing) can make $P^{(\text{HD})}(x; |\psi_{1,2}\rangle)$ approach $P^{(\text{min})}(|\psi_1\rangle, |\psi_2\rangle)$ as much as possible with respect to other combinations.

2.1 Discrimination between Coherent States

In the case of coherent state communication our states are defined as

$$|\psi_k\rangle = \hat{D}\left[(-1)^k a\right] |0\rangle, \quad k = 1, 2 \quad (22)$$

where $a \in \mathbb{R}$ and $k = 1, 2$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P_{\text{CS}}^{(\text{min})}(N) = \frac{1}{2} - \frac{1}{2} \sqrt{1 - \exp(-4N)}, \quad (23)$$

where N is the average number of photons in the coherent states. The homodyne detection probability distribution becomes

$$P_{\text{CS}}^{(\text{HD})}(x; |\pm a\rangle) = \frac{1}{2} \text{erfc}\left(\sqrt{2N}\right). \quad (24)$$

2.2 Discrimination between DSS

In the case of communication with DSS, our states are defined as

$$|\psi_k\rangle = \hat{D}\left[(-1)^k a\right] \hat{S}(r) |0\rangle, \quad k = 1, 2 \quad (25)$$

where $z \in \mathbb{R}$ and $k = 1, 2$ is selected with equal probability. The minimum error probability (Helstrom Bound) is given by

$$P^{(\text{min})}(N) = \frac{1}{2} - \frac{1}{2} \sqrt{1 - \exp(-4Ne^{-2r})}, \quad (26)$$

where N is the average number of photons in the DSS

$$N = |a|^2 + \sinh^2(r). \quad (27)$$

We may define the squeezing fraction $\beta \equiv \sinh^2(r)/N$, and express the relevant equations with respect to that. The minimum error probability (Helstrom Bound) is given by

$$P_{DSS}^{(\min)}(N, \beta) = \frac{1}{2} - \frac{1}{2} \sqrt{1 - \exp \left[-4N(1 - \beta) \times \left(1 + 2N\beta + 2\sqrt{N\beta(1 + N\beta)} \right) \right]}, \quad (28)$$

The homodyne detection probability distribution becomes

$$P_{DSS}^{(\text{HD})}(x; |\pm a\rangle) = \frac{1}{2} \text{erfc} \left(\sqrt{2} \frac{\alpha}{\Sigma} \right), \quad (29)$$

where

$$\alpha = \sqrt{N(1 - \beta)}, \quad \Sigma = \left(\sqrt{N\beta} + \sqrt{1 + N\beta} \right) \quad (30)$$

2.3 Which is easier to discriminate, Coherent states or DSS?

For a certain fraction of squeezing, DSS can outperform Coherent states. More specifically, for a fixed number of photons $\tilde{N}$, DSS have a lower error probability if β is below a threshold β_{th}

$$P_{\text{err}}^{DSS}(\beta | \tilde{N}) \leq P_{\text{err}}^{CS}(\tilde{N}) \quad \text{if } \beta \leq \beta_{\text{th}}(\tilde{N}), \quad (31)$$

where

$$\beta_{\text{th}}(\tilde{N}) = \frac{4\tilde{N}}{4\tilde{N} + 1}. \quad (32)$$

Additionally, for a fixed number of photons $\tilde{N}$, there exists an optimal fraction of squeezing $\beta_{\text{opt}} \in [0, \beta_{\text{th}}]$ for which the error probability is minimized

$$\left. \frac{\partial P_{\text{err}}^{DSS}}{\partial \beta} \right|_{\beta_{\text{opt}}, \tilde{N}} = 0 \Rightarrow \beta_{\text{opt}} = \frac{\tilde{N}}{2\tilde{N} + 1}. \quad (33)$$

Summarizing, for a fixed number of photons $\tilde{N}$, DSS will always outperform Coherent states if the fraction of squeezing is lower than $\beta_{\text{th}}(\tilde{N})$. Moreover, the error probability will be minimized if the fraction of squeezing is equal to $\beta_{\text{opt}}(\tilde{N}) < \beta_{\text{th}}(\tilde{N})$.

3 Discrimination between Quantum States under phase diffusion in a communication channel